KPI LIST;

1) Store Waste Rate= (Store Waste Crates / Delivered Crates) × 100

= (1.787/47.026)x100≈ %3.8

2) Depot Waste Rate= (Depot Waste Crates / Produced Crates) × 100

=(682/47.480)x100 ≈ %1.4

3) Sold Over Quality Limit= (Sold Over Quality Limit Crates/Total Sold Crates)x100

=(545/45.263)x100 ≈ %1,204

4)Lost Salest Rate=(Lost Sales Crates/Total Demand Crates)x 100

=(142/45.405)x100 ≈ %0.3

5) Transport Efficiency & Logistics Cost

Total Transportation Cost=(Fixed Cost(TRY/day))+ Distance covered(km)+ Cost per km(TRY/km)

= ∑*km=37.648 km*
= ∑*cost=1.193.010 TRY*
= *Average Daily Distance:37.648/183=205,727 km/day*
=*Average Daily Cost: 1.193.010/183=6.519,18 TRY/day*
= *Unit Transportation Cost: 1.193.010/37.648=31,689 TRY/km*

6) Late Delivery Rate = (∑*Late Stops)/(*∑*Total Stops)x100*

Local Stop: 183 day x 15/day=2745 stop
District Stop: 78 day x 5/day=390 stop
Total Stop:3135 stop
Total Number of Delayed Stop: 3135x0.031=97 stop

=Late Delivery Rate = (97/3.135)x100=3,0941%

Critical Route;R3 Late Delivery Rate
=(98/390)x100=25,128%

The delay rate across the general network is around 3%. However, on the R3 route, it is at 25%. One out of every four deliveries is late.

Analyse the baseline KPIs by month and by store;

1) **Impact of Extreme Summer Temperatures:** During August, ambient temperatures reach up to ~43 °C. Milk crates exposed during ambient transfer stages—specifically the unmonitored loading dock (~15 min) and retail receiving floor (0.5–2.5 h)—experience accelerated bacterial growth. Consequently, milk sold exceeding the microbiological safety limit (*20,000 cfu/mL*20,000 cfu/mL) reaches an annual peak of 1.9% of total retail sales in August..

2) **The Peak of Market Waste:** Market waste reached 4.2% in August and September, rising notably above the annual average of 3.8%**.**

3) **Spatial & Asset Analysis: Baseline Losses by Store**

3.1) Grocery Stores with Poor Refrigeration Units: Even if the milk reaches the retail outlet safely, it is often stored in inadequate refrigeration units at temperatures around 6–8 °C; as a result, its remaining shelf life (RSL) is depleted much faster than the date indicated on the label, and the bacterial count exceeds the 20,000 cfu/mL threshold well before the expiration date (use-by date) is reached.

3.2) District Markets (Route D01–D05 / R3): 25% of the stops are experiencing delays, and the 07:00–12:00 time window is being exceeded. Due to long transit times and the frequent opening and closing of vehicle doors in each district, the milk warms up significantly during transport; consequently, by the time the product reaches the shelves in district retail outlets, its remaining shelf life (RSL) is already nearing its end.

Proposed solutions:
1) Stores with poor refrigeration units or high sales turnover should be supplied with fresher milk—that is, milk with a longer remaining shelf life (RSL).
2) The time that milk—destined for long-distance district markets—spends in transit must be taken into account, and the R3 route should be replanned.

Gantt chart:

Task / Deliverable	Lead Role	Timeline	W1	W2	W3 (M1)
1. Dataset Setup & Role Assignment	All	Week 1	**Done**		
2. Baseline Exploration & Initial Models	All Leads	Week 2		**Done**	
3. Milestone 1: Project Proposal & Design (M1)	All Leads	Week 3			**★ M1**